

Functional Central Limit Theorem

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FCLT

- The **Functional Central Limit Theorem** (FCLT) describes the limiting behavior as $T \rightarrow \infty$ of a scaled cumulated sum

$$\hat{S}_z(\tau) \equiv \frac{1}{\sqrt{T}} \sum_{t=1}^{\tau T} z_t, \quad \tau = \frac{1}{T}, \frac{2}{T}, \dots, 1,$$

viewed as a *function* $\hat{S}_z: [0, 1] \rightarrow \mathbb{R}$.

- The FCLT is a useful theoretical tool in many contexts:
 - ▶ Testing for structural breaks.
 - ▶ Unit root models.
 - ▶ Fixed- b HAR inference. See Kiefer & Vogelsang (2002, 2005).
 - ▶ Etc.

Scaled random walk

$$\hat{S}_z(\tau) = \frac{1}{\sqrt{\tau}} \sum_{t=1}^{\tau} z_t, \quad \tau = \frac{1}{T}, \frac{2}{T}, \dots, 1$$

- To start, assume that

$$z_t \stackrel{iid}{\sim} N(0, 1).$$

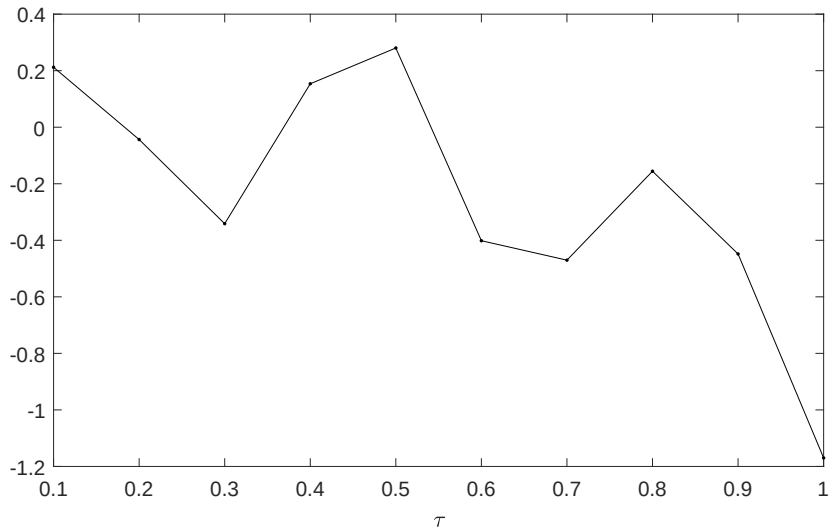
- This is a **scaled random walk**: sum of i.i.d. (here: Gaussian) increments z_t .
- Then $E[\hat{S}_z(\tau)] = 0$ for all $\tau = \frac{k}{T} \in [0, 1]$, $k \in \mathbb{N}$, and

$$\hat{S}_z(\tau_2) - \hat{S}_z(\tau_1) \sim N(0, \tau_2 - \tau_1) \quad \text{for all } \tau_1 < \tau_2,$$

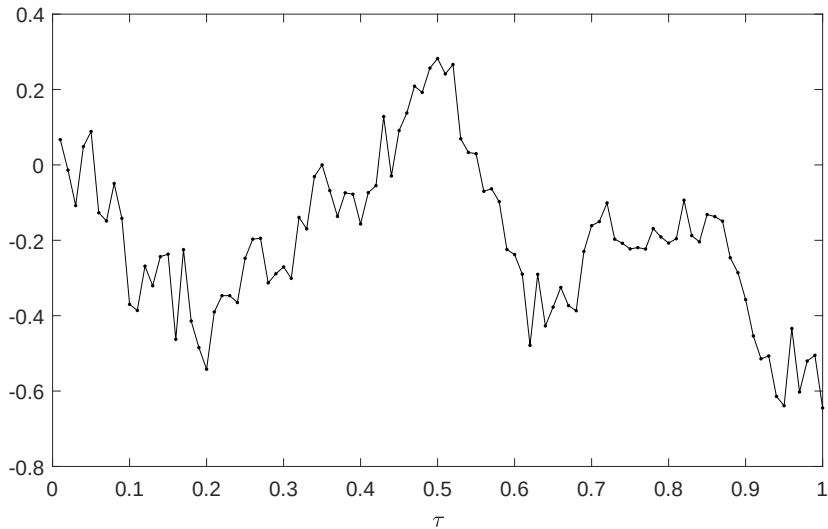
$$(\hat{S}_z(\tau_4) - \hat{S}_z(\tau_3)) \perp\!\!\!\perp (\hat{S}_z(\tau_2) - \hat{S}_z(\tau_1)) \quad \text{for all } \tau_1 < \tau_2 < \tau_3 < \tau_4.$$

- The fact that $\text{Var}[\hat{S}_z(1)] = 1$ motivates our choice of the scaling factor $\frac{1}{\sqrt{\tau}}$ in the definition of $\hat{S}_z(\cdot)$.

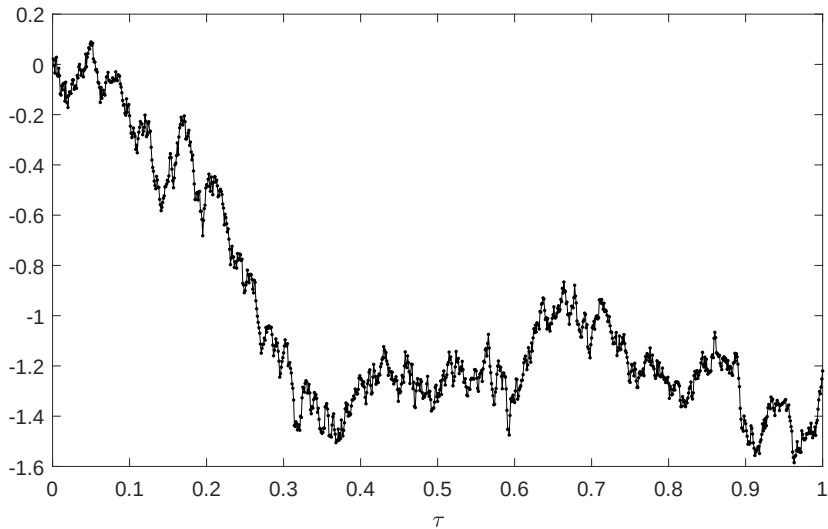
Scaled random walk, $T = 10$



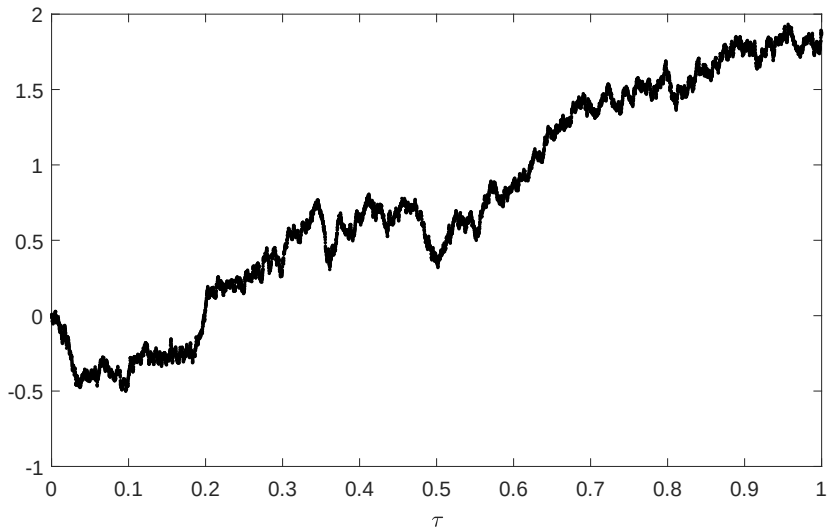
Scaled random walk, $T = 100$



Scaled random walk, $T = 1000$



Scaled random walk, $T = 10000$



Scaled random walk (cont.)

- As $T \rightarrow \infty$, the plot of the random walk converges to a sample path of **standard Brownian motion** (also called a **Wiener process**).

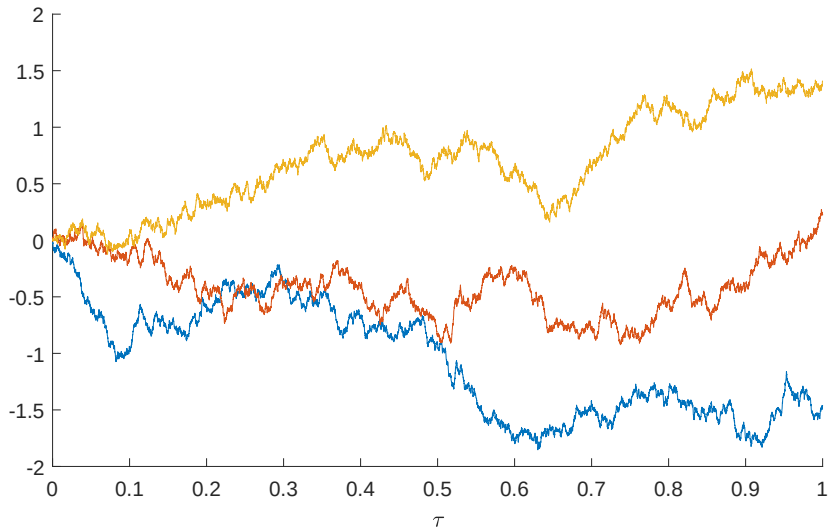
Definition

Standard Brownian motion (BM) is a random process $B(\cdot)$ defined on the unit interval $[0, 1]$ satisfying:

- i) $B(0) = 0$.
- ii) $B(\tau_2) - B(\tau_1) \sim N(0, \tau_2 - \tau_1)$ for all $0 \leq \tau_1 < \tau_2 \leq 1$.
- iii) $(B(\tau_4) - B(\tau_3)) \perp\!\!\!\perp (B(\tau_2) - B(\tau_1))$ for all $0 \leq \tau_1 < \tau_2 < \tau_3 < \tau_4 \leq 1$.
- iv) Sample paths are continuous almost surely.

- See Davidson (1994), ch. 27, for a proof that BM exists.
- Sample paths of BM are nowhere differentiable almost surely!

Standard Brownian motion sample paths



Weak convergence of functions

- The previous slides suggest that the scaled Gaussian random walk

$$\hat{S}_z(\tau) \equiv \frac{1}{\sqrt{T}} \sum_{t=1}^{\tau T} z_t, \quad z_t \stackrel{iid}{\sim} N(0, 1),$$

converges in distribution to standard BM, as a *function* defined on the unit interval $[0, 1]$.

- To define $\hat{S}_z(\tau)$ on the entire interval $[0, 1]$, we set $\hat{S}_z(0) = 0$ and linearly interpolate between the points $0, \frac{1}{T}, \frac{2}{T}, \dots, 1$.
- For large T , this linear interpolation will have negligible effect, since the points $\frac{1}{T}, \frac{2}{T}, \dots, 1$ will be very finely spaced.
- We now define convergence in distribution for a random function $\hat{S}_z(\cdot)$ defined on $[0, 1]$.

Weak convergence of functions (cont.)

Definition

Let $\mathcal{C}[0, 1]$ denote the space of continuous functions on $[0, 1]$. A random function $\hat{S} = \hat{S}(\cdot)$ **converges weakly** to $S = S(\cdot)$ if both take values in $\mathcal{C}[0, 1]$ almost surely and

$$E[h(\hat{S})] \rightarrow E[h(S)] \quad \text{as } T \rightarrow \infty,$$

for every function $h: \mathcal{C}[0, 1] \rightarrow \mathbb{R}$ that is bounded and uniformly cts wrt. the supremum norm $\|s\|_{\text{sup}} \equiv \sup_{\tau \in [0, 1]} |s(\tau)|$.

We then write $\hat{S}(\cdot) \Rightarrow S(\cdot)$.

- Examples of $h: \mathcal{C}[0, 1] \rightarrow \mathbb{R}$:

$$h(s) = \int_0^1 s(\tau) d\tau, \quad h(s) = s(1), \quad h(s) = \sup_{\tau \in [0, 1]} s(\tau).$$

Weak convergence of functions: sufficient conditions

Proposition (Andrews, 1994, p. 2251)

We have $\hat{S}(\cdot) \Rightarrow S(\cdot)$ if the following two conditions hold:

i) For any k and any $0 \leq \tau_1 < \dots < \tau_k \leq 1$,

$$(\hat{S}(\tau_1), \dots, \hat{S}(\tau_k))' \xrightarrow{d} (S(\tau_1), \dots, S(\tau_k))' \quad \text{as } T \rightarrow \infty.$$

ii) For any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\limsup_{T \rightarrow \infty} P \left(\sup_{\tau_1, \tau_2: |\tau_1 - \tau_2| < \delta} |\hat{S}(\tau_1) - \hat{S}(\tau_2)| > \varepsilon \right) < \varepsilon.$$

- Condition (i), finite-dim. convergence, is often verified with a CLT.
- Condition (ii) is called **stochastic equicontinuity**. Requires $\hat{S}(\cdot)$ to not vary drastically across nearby τ , with prob. $\rightarrow 1$.

Functional central limit theorem

- Consider again

$$\hat{S}_z(\tau) \equiv \frac{1}{\sqrt{T}} \sum_{t=1}^{\tau T} z_t, \quad z_t \stackrel{iid}{\sim} N(0, 1),$$

with $\hat{S}_z(\tau)$ defined by linear interpolation on the rest of $[0, 1]$. Then $\hat{S}_z(\cdot) \in \mathcal{C}[0, 1]$ almost surely.

- Donsker's theorem** says that

$$\hat{S}_z(\cdot) \Rightarrow B(\cdot),$$

where the limit is a standard BM.

- The result in fact holds for more general innovation processes $\{z_t\}$.
- This last fact is called the **functional central limit theorem** (FCLT) or **weak invariance principle** (invariance to the distr. of z_t).

FCLT for martingale difference sequences

Proposition (Davidson, 1994, Thm. 27.14)

Let $\{z_t\}$ be a univariate strictly stationary MDS wrt. $\mathcal{F}_t$. Assume

$$\frac{1}{T} \sum_{t=1}^T z_t^2 \xrightarrow{P} \sigma^2 > 0 \quad \text{as } T \rightarrow \infty.$$

Define

$$\hat{S}_z(\tau) \equiv \frac{1}{\sqrt{T}} \sum_{t=1}^{\tau T} z_t \quad \text{for } \tau = \frac{k}{T} \in (0, 1], \quad k \in \mathbb{N},$$

and linearly interpolate $\hat{S}_z(\tau)$ for other $\tau \in [0, 1]$. Then

$$\hat{S}_z(\cdot) \Rightarrow \sigma B(\cdot) \quad \text{as } T \rightarrow \infty.$$

Here $B(\cdot)$ is standard Brownian motion on $[0, 1]$.

Partial sum process

- The process $\hat{S}_z(\tau) \equiv \frac{1}{\sqrt{T}} \sum_{t=1}^{\tau T} z_t$ is called a **partial sum process**.
- It's annoying and unnecessary to deal with the asymptotically negligible linear interpolation at non-fractional $\tau \in [0, 1]$.
- Instead, we from now on define

$$\hat{S}_z(\tau) \equiv \frac{1}{\sqrt{T}} \sum_{t=1}^{[\tau T]} z_t, \quad \tau \in [0, 1],$$

where $[x]$ is the integer part of x . Then $\hat{S}_z(\tau)$ is a step function.

- $\hat{S}_z(\cdot)$ is not in $\mathcal{C}[0, 1]$ anymore. It lies in the space $\mathcal{D}[0, 1]$ of **càdlàg** functions that are cts from the right and have limits from the left.
- Weak convergence on $\mathcal{D}[0, 1]$ is defined analogously to $\mathcal{C}[0, 1]$, with a few measure-theoretic complications. The distinction does not matter for anything practical. See Davidson (1994), ch. 28.

FCLT for serially dependent processes

- To obtain a FCLT for a univariate linear process $y_t = \Theta(L)z_t$, where z_t is a MDS w. var. σ_z^2 , we apply the Phillips-Solo device:

$$\begin{aligned}\hat{S}_y(\tau) &\equiv T^{-1/2} \sum_{t=1}^{[\tau T]} y_t = T^{-1/2} \sum_{t=1}^{[\tau T]} \{\Theta(1)z_t + \eta_t - \eta_{t-1}\} \\ &= \Theta(1)T^{-1/2} \sum_{t=1}^{[\tau T]} z_t + T^{-1/2}(\eta_{[\tau T]} - \eta_0) = \Theta(1)\hat{S}_z(\tau) + o_p(1) \\ &\Rightarrow \Theta(1)\sigma_z B(\tau).\end{aligned}$$

- There also exist FCLTs for more general dependent processes (e.g., near-epoch dependent). See Davidson (1994), ch. 29.
- General lesson: If the n -dim. time series $\{y_t\}$ has LRV matrix Ω , then

$$\hat{S}_y(\cdot) \Rightarrow \Omega^{1/2} B_n(\cdot),$$

where $\Omega^{1/2}\Omega^{1/2'} = \Omega$ (e.g., Cholesky decomposition), and $B_n(\cdot)$ is **n -dimensional standard BM** (vector of indep. standard BMs).

Continuous mapping theorem

Proposition (Davidson, 1994, Prop. 26.13)

Let $g: \mathcal{D}[0, 1] \rightarrow \mathbb{R}^k$ be a continuous function wrt. the supremum norm. If $\hat{S}(\cdot) \Rightarrow S(\cdot)$, then

$$g(\hat{S}) \xrightarrow{d} g(S).$$

- Example: Let $\hat{S}_z(\tau)$ be a partial sum process w. MDS innovations of variance σ^2 . Then

$$\int_0^1 \hat{S}_z(\tau) d\tau \xrightarrow{d} \sigma \int_0^1 B(\tau) d\tau \quad \text{as } T \rightarrow \infty,$$

where $B(\cdot)$ is standard BM. **Exercise:** Show that this map is cts.